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COLREG compliant trajectory planning for autonomous vessels in areas with static and dynamic obstacles

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***Abstract*— The development of autonomous vessels is a topic of great interest in the maritime industry. Generating trajectories for vessels over long time horizons while respecting multiple sets of rules, is a complicated task. In this contribution, a procedure to generate trajectories that comply with the rules of preventing collisions at sea (COLREGs) is presented. The approach is based on the combination of three methods, namely the rapidly exploring random tree (RRT) algorithm, velocity obstacles (VOs), and motion primitives (MPs). In this approach, the dynamic situation is analyzed and the corresponding COLREG rule is enforced by selecting only valid, i.e., COLREG conform, MPs from a database. This database is created by solving a boundary value problem (BVP) for a set of initial and final conditions. The approach is capable of respecting static and dynamic obstacles and can be used for a variety of vessel classes and time horizons.**

I. INTRODUCTION

The development of autonomous vessels is a topic of great interest in the maritime industry, motivating further development of high performing assistance, or even completely autonomous systems. Analyzing the dynamic environment at all times and suggesting alternative routes that prevent complex COLREG scenarios from happening in the first place, can be a huge factor in increasing the overall safety of maritime traffic. Additionally, such measures can also lead to increased fuel efficiency, as unnecessary course changes can be avoided. While this contribution is motivated from applications in the maritime sector, the proposed approach can be applied in various other domains including autonomous driving or autonomous robots.

The main challenge consists in finding the optimal state and input trajectory for a vessel navigating a predefined route in a highly dynamic environment. While many search algorithms are probabilistically complete, meaning they will find a solution in finite time, the speed of convergence is not guaranteed. The complexity of accurately modeling a highly dynamic environment, coupled with the necessity to adhere to a multitude of regulatory constraints, significantly increases the computational burden. This restrains the practical applicability of many approaches, since computational time is limited and motivates the need for fast and efficient extensions of existing algorithms, to further improve computation time and extend the set of applicable scenarios.

Finding solutions to the trajectory generation problem in complex situations is a recent and ubiquitous topic in the maritime sector. Methods range from model predictive control (MPC) approaches [8], [15], [19] to methods based on search algorithms [1], [2], [4]–[7], [21], where multi-layered approaches are also common. Search algorithms are often preferred due to their flexibility and adaptability in various scenarios. Current contributions combine only some of the relevant aspects of trajectory generation, like the combination of RRT with MP, RRT with VOs, or MP with VOs. Trajectory generation is such a complex problem that all relevant aspects need to be considered.

For this reason this contribution proposes in a unified approach. It is shown how COLREG information can efficiently be respected using MPs in combination with VOs. MPs, which are a priori solutions of vessel maneuvers, lower the numerical burden significantly. By comparing the whole set of MPs against the VO of the target to reach COLREG conformity, only valid maneuvers remain. The resulting reduced set of MPs can be evaluated using a cost function to retrieve the desired MP for the current node. Iteratively searching for the best possible MP on a reduced set of the database using a sampling based search algorithm efficiently leads to a global COLREG conform and collision free trajectory. Additionally this contribution introduces the concept of the swept hull for the avoidance of static obstacles into the context of sampling based trajectory generation. It is shown, how it seamlessly integrates into the proposed approach and lowers computational effort.

This paper is organized as follows. First, a summary of COLREGs is given in Sec. II, followed by the presentation of the underlying dynamic model of the vessel in Sec. III. Afterwards, the unified procedure to solve the trajectory generation problem is explained in Sec. IV. Within this section, the combination of RRT, MP and VO is shown. Subsequently, the simulation results are discussed in Sec. V and an outlook is given in Sec. VI.

II. SUMMARY OF COLREGS

Regulations regarding the prevention of collisions of marine vessels at sea are described by the International Maritime Organization (IMO) [13]. Within the given set of rules, paragraphs 13-17 are of particular interest, as they describe the behavior of vessels in sight of each other. The rules describe the behavior in certain configurations. Overtaking, crossing and head-on are the three main configurations, where an avoiding action by the ownship has to be taken. Other configurations, being-overtaken and crossing-stand-on, are also described, but here the ownship is not required to take an avoiding action and is required to keep course

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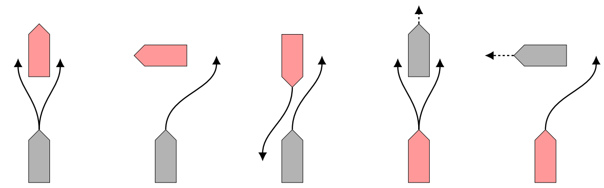

Fig. 1: Schematic representation of the COLREG states with the ownship colored gray and the target red: Overtaking, crossing, head on, being-overtaken and crossing-stand-on, from left to right.

and speed. All these configurations are presented in Fig. 1. Overshadowing all rules, of course, is the obligation to prevent collision at all times. So, in an emergency situation, where the target is not responding adequately, one is required to take an avoiding action. The exact distances (DCPA) and time to closest approach (TCPA) are not specified in these documents, as they vary depending on the types of vessels that encounter each other and the external circumstances. This represents a degree of freedom that has to be analyzed from common practice and values have to be set accordingly. The correct selection and translation of the COLREG rule into the mathematical counterpart is crucial for the correct behavior of the ownship, but not in the scope of this contribution.

III. Dynamic model

Respecting the set of rules imposed by the COLREGs is further complicated by the need to simultaneously address the vessel dynamics. It is not sufficient to just compute a collision-free path, because it is constraining the vessel's maneuverability. Model based approaches solve this problem, as the vessel dynamics are included in the planning procedure. These dynamic models exist in different levels of abstraction. In this contribution a Nomoto-class vessel dynamical model, see [10], is used.

The state of the vessel in the North-East-Down (NED) frame, which is in this case assumed to be a inertial frame, can be divided into

$$\boldsymbol{x} = [\boldsymbol{\eta}^\top, \boldsymbol{\nu}^\top]^\top \in \mathbb{R}^n, \tag{1}$$

where n is the number of states, $\boldsymbol{\eta}$ denotes the pose in the NED-frame and $\boldsymbol{\nu}$ the velocities in the body-fixed frame of the vessel. To arrive at this description, first, the kinematic relations, as in Fig. 2, for a vessel in the NED-frame are analyzed. Since a Nomoto-class vessel is desired, the yaw dynamics are approximated using a PT1 system. This approach simplifies the process, requiring only the identification of two parameters, see [10]. Additionally, using Nomoto models, the assumption is made that the vessel is moving with constant surge velocity $u = u_d$, which is a reasonable assumption for the autopilot model, hence the sway can be set to $v = v_d = 0$. Obtaining the velocity vector $\boldsymbol{\nu}_d = [u, v, r]^\top = [u_d, 0, r]^\top$, with constant u_d, the resulting state vector reads

$$\boldsymbol{x} = [x, y, \psi, r]^\top \in \mathbb{R}^4, \tag{2}$$

where x is the position in North direction, y in East direction, ψ the yaw angle (heading), and r the yaw-rate. Furthermore, the kinematic relation between velocities in the NED-frame and the body-fixed frame of the vessel is given by

$$\dot{\boldsymbol{\eta}} = R(\psi)\boldsymbol{\nu}_d = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \boldsymbol{\nu}_d = \begin{bmatrix} \cos(\psi)u_d \\ \sin(\psi)u_d \\ r \end{bmatrix}. \tag{3}$$

The PT1 system for the yaw-rate is introduced as

$$\dot{r} = \frac{1}{T}(K\delta - r), \tag{4}$$

where the input δ is the rudder angle and the parameters K, T are made dimensionless by introducing the transformation $K = \frac{u}{L_{pp}}K'$ and $T = \frac{L_{pp}}{u}T'$. Here, L_{pp} is the length of the vessel. The full state space representation of the model is then given by

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \delta) = \begin{bmatrix} \cos(\psi)u_d \\ \sin(\psi)u_d \\ r \\ -\frac{u_d}{L_{pp}}(1/T')r + (\frac{u_d}{L_{pp}})^2(K'/T')\delta \end{bmatrix}. \tag{5}$$

The equations of motion (5) have to be respected in order to derive feasible trajectories.

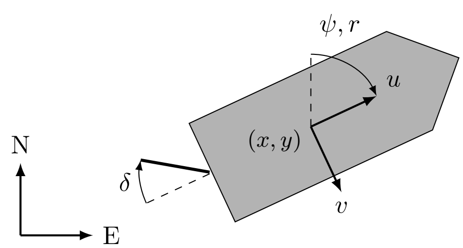

Fig. 2: Kinematics of the vessel, with inertial (NED)-frame and body-fixed frame. The rudder angle is denoted by δ.

IV. COLREG conformal trajectory generation

Generating a global trajectory for the vessel can be seen as an optimization problem. Solving this dynamic problem using techniques from optimal control theory leads to a long time horizon and therefore a large number of discretization steps. As a consequence the solution of the resulting optimal control problem becomes challenging and is further complicated by the need to consider complementary rules, in particular the COLREGs. For this reason, other methods are of interest. Instead of solving the problem directly, search algorithms can be used. In this case the RRT is used to search in the state space using random sampling. As the procedure is still required to respect the vessel's dynamics, in each iteration of the RRT a boundary value problem (BVP) has to be solved to connect nodes of the tree. This introduces high computational effort, as the tree consists of a huge amount of nodes. Here, MPs are exploited to connect two nodes of the RRT. As an MP is an abstraction of the solution of the BVP connecting the nodes, the system dynamics are respected at all time, while the numeric complexity is reduced. To address the COLREG rules, the VO of the target is calculated and manipulated according to the corresponding rule. The individual parts of the algorithm are explained below.

A. Rapidly-exploring Random Trees

There exists a variety of different search algorithms suitable for the foundation of the proposed algorithm. While graph-based algorithms like A* are more efficient in finding the optimal solution in undisturbed environments, they suffer in situations with multiple obstacles. Lattice-based planners, like in [1], are also a viable option, but they require the database to create a state lattice, which is not the most natural description of the problem. While the approach in general is independent of the underlying search algorithm, sampling-based algorithms, like the RRT [17], bring some benefits. Based on their randomness they are able to adapt to many scenarios. Without going into detail, the RRT is a sampling-based algorithm having a tree as a growing data structure. The tree consists of nodes and edges. The base search algorithm has the task to grow the tree by iteratively finding random nodes and connecting them by creating the edge. When taking a random sample from the state space and connecting it to the closest node of the tree, a BVP has to be solved. This is due to the fact that we have to satisfy the kinodynamic constraints imposed by the dynamic model (5). Computing the BVPs can become challenging.

B. Motion Primitives

To reduce the numeric burden, instead of solving the BVPs online, a database of pre-computed trajectories is used. Using the equations of motion (5) a set of pre-defined trajectories are generated, which are referred to as MPs. This MP from the database is selected and subsequently transformed to the nearest neighbor node. The final state of the MP then constitutes the new node. This technique is often used in robotic applications, where an MP can vary in abstraction from just a small movement of a joint to a full sequence of actions. MPs will be considered to represent parts of the global trajectory, as they represent the edges of the tree. It is important to node that this procedure converts the solution of the optimization problem of finding a COLREG conformal trajectory into a sub-optimal solution, but vastly lowers the numeric complexity. The assumption is made that minor changes in heading angles are not crucial for the overall performance.

As the MPs represent the edges of the tree, MPs are required to be continuous when being connected. This implies that the velocities of the trajectories of the MPs must be the same at the start and the end. As previously mentioned, the speed over ground (SOG) u_d is set to be constant over time. The implication is that the yaw-rate r, as in (2), at start and end is $r_0 = r_f = 0$ [3]. The same applies for the input $\delta_0 = \delta_f = 0$. This can be seen as a desired behavior for the ship and integrates well with the underlying Nomoto model. Using these assumptions, an MP always describes a full maneuver, which is a course-change in this contribution. If the time horizon of the MPs is increased, the final solution will consist of less MPs as with shorter time horizons. This directly influences the computational burden, as most of the computation time is spent checking the MPs for validity. But it also influences the quality of the solution, as longer MPs tend to be more abstract and more detailed solutions may not be possible. To arrive at a set of MPs, which is further referred to as the database, a generic BVP$_i$ is solved for a set of final conditions given fixed initial condition. The BVP$_i$ results from the constrained optimal control problem

$$\min_{\boldsymbol{x},\delta} \int_{t_0}^{t_{f,i}} \frac{1}{2}\delta^2 \, dt \tag{6a}$$
$$\text{s.t.} \quad \dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x},\delta), \quad t > t_0 \tag{6b}$$
$$\boldsymbol{x}(0) = \boldsymbol{x}_0 \tag{6c}$$
$$\boldsymbol{x}(t_{f,i}) = \boldsymbol{x}_{f,i} \tag{6d}$$
$$\delta_{min} \leq \delta \leq \delta_{max}, \tag{6e}$$

which includes a control-action minimizing cost-function $l(\delta) = \frac{1}{2}\delta^2$, with the rudder angle bounded to $\delta \in [\delta_{min}, \delta_{max}]$. It is necessary to impose that $\boldsymbol{\nu}_f = \boldsymbol{\nu}_0$, as initial and terminal velocities are not free. The pose is always assigned as the origin $\boldsymbol{\eta}_0 = [0,0,0]^\top$. Therefore, the free parameters are $\boldsymbol{\eta}_{f,i} = [x_{f,i}, y_{f,i}, \psi_{f,i}]^\top$, which now used to define the database. Defining the position $(x_{f,i}, y_{f,i})$ is not necessary, but can improve solver convergence. The BVP$_i$ (6) can consequently be solved using numerical techniques. In this contribution, a multiple-shooting approach is used, where the control inputs, as well as the state variables are discretized.

Solving (6) for a vessel of length $L_{pp} = 60\,\mathrm{m}$, at first for a final time $t_f = 30\,\mathrm{s}$ and for a set of final heading angles $\psi_f \in [-25°, 25°]$ with a resolution of $\Delta\psi_{f,i} = 5°$ and secondly for a final time of $t_f = 60\,\mathrm{s}$ and $\psi_f \in [-180°, 180°]$, results in a usable database, which is depicted in Fig. 3. Here, starboard maneuvers are shown green and portside maneuvers are shown red. The prescribed SOG is $u_d = 5\,\mathrm{kn} = 2.57\,\mathrm{m\,s^{-1}}$. The achieved database, in which each MP is a combination of $(\boldsymbol{x}^*(t), \delta^*(t))$, has the important property of being invariant to transformations in space. The

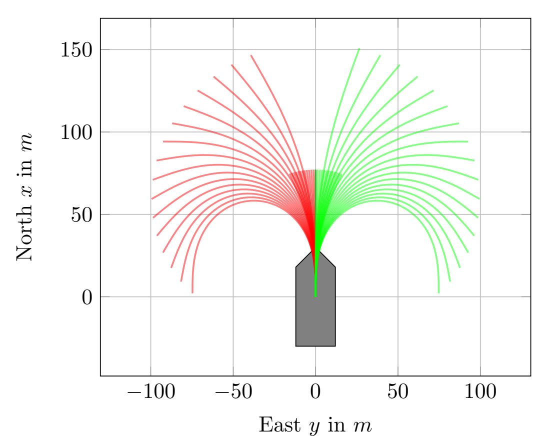

Fig. 3: Trajectories of the states (x, y) in the NED-frame for a set of final heading angles with a $\Delta\psi_{f,i} = 5°$ discretization.

input trajectory δ^* and velocities in the body-fixed frame generally are independent on the pose in space.

C. Obstacle avoidance

Obstacle avoidance is one, if not the most crucial part of generating trajectories in the context of autonomous vessels. Two categories can be distinguished, i.e. static and dynamic obstacle avoidance. In theory, similar approaches for both kinds of obstacles can be used. According to the COLREGs, the avoidance of moving obstacles, is required to be early, thorough and clearly recognizable.

1) Static obstacle avoidance: In recent years multiple approaches have been explored. Subsequently, methods are considered that exploit the description of obstacles as convex polygons, see, e.g., [12]. These approaches can make use of well known and fast algorithms, like GJK [11]. This algorithm iteratively checks, if two convex polygons are overlapping.
Checking for a collision between the ownship and a static obstacle is done by checking the polygon of the ownship against the convex polygon of the obstacle. For an MP, it is necessary to use the GJK algorithm at all discretization steps. However, this is not efficient when the number of discretization steps is large, i.e., when MPs tend to be long. As a simplification, the MPs are checked for collision with the convex hull of the solution path. This approach is closely related to the idea of swept volumes, see [18], [20]. Using rather common and simple algorithms, the convex hull of an MP can be calculated before run-time and then transformed to the pose of interest. Here, the *gift-wrapping* [14] algorithm is used, whose details are not in the scope of this contribution. As the calculations are done a-priori the efficiency of the used algorithm is not of great importance. With the calculated swept hull, only one iteration of the GJK algorithm is needed to check for collisions, which is a significant improvement. The impact is depicted in Fig. 4. Here, the left figure describes an MP consisting of a large amount of time steps and the huge amount of associated polygons to be collision checked. In contrast, in the right figure only one polygon (orange) is seen that has to be checked using a single run of the GJK algorithm. It becomes clear that an error is introduced, as the swept hull is always larger than the area covered by the single polygons. The error grows with the desired change in course, but is negligible compared to the safety margins of objects.

2) Dynamic obstacle avoidance: In contrast to static Obstacles, a common way to treat dynamic obstacles is given by so-called VOs. Based on [9], applications in the maritime sector are starting with [16]. These exploit the idea of transforming the problem from the inertial NED-frame into the velocity space, which is suitable for moving obstacles. So, instead of defining positions that are to be avoided, velocities leading to collisions are detected and avoided. Beginning with the definition of the set of points describing the ownship ($\mathcal{O}$) and the target of interest ($\mathcal{T}$), corresponding positions ($\boldsymbol{p}_{\mathcal{O}}$, $\boldsymbol{p}_{\mathcal{T}}$) and velocities ($\boldsymbol{v}_{\mathcal{O}}$, $\boldsymbol{v}_{\mathcal{T}}$) are defined. The whole procedure depends on the linear prediction of the path taken by the target vessel in the form of

$$\lambda(\boldsymbol{p}, \boldsymbol{v}) = \{\boldsymbol{p} + t\boldsymbol{v} \mid t \geq 0\}, \tag{7}$$

where $\boldsymbol{p} = [x, y]^\top$ is the current position of the target and $\boldsymbol{v}$ the current inertial velocity. With this, the VO, from the perspective of the ownship to a target vessel, can be defined as

$$VO_{\mathcal{T}}^{\mathcal{O}}(\boldsymbol{v}_{\mathcal{T}}) = \{\boldsymbol{v}_{\mathcal{O}} \mid \lambda(\boldsymbol{p}_{\mathcal{O}}, \boldsymbol{v}_{\mathcal{O}} - \boldsymbol{v}_{\mathcal{T}}) \cap (\mathcal{T} \oplus -\mathcal{O}) \neq \emptyset\}, \tag{8}$$

where the Minkowski sum is defined as

$$O \oplus T = \{\boldsymbol{p}_O + \boldsymbol{p}_T \mid \boldsymbol{p}_O \in \mathcal{O}, \boldsymbol{p}_T \in \mathcal{T}\}. \tag{9}$$

The VO describes the set of inertial velocities that lead to a collision, based on $(\mathcal{O}, \mathcal{T})$ and a linear prediction of both participants. In Fig. 5 the setting of a vessel-to-vessel encounter is illustrated. Here, the NED-frame, as well as the velocities in the NED-frame are shown, where in contrast to $\boldsymbol{\nu}$, the inertial velocities are described by $\boldsymbol{v}$. As can be seen in Fig. 5, the VO and collision cone (CC) are closely related. In fact, the CC is the same cone as VO, but the CC is not translated by the inertial target velocity $\boldsymbol{v}_{\mathcal{T}}$. The CC is defined by the cone of two halfplanes, which define the normal vectors ($\boldsymbol{n}_{\mathcal{CC},l}$, $\boldsymbol{n}_{\mathcal{CC},r}$). These normal vectors can be used to formulate the obstacle avoidance rules. In general the rules can be described by

$$\boldsymbol{v}_{\mathcal{O}}^* \notin \mathcal{VO}_{\mathcal{T}}^{\mathcal{O}} \quad \vee \quad (\boldsymbol{v}_{\mathcal{O}}^* - \boldsymbol{v}_{\mathcal{T}}) \notin \mathcal{CC}_{\mathcal{T}}^{\mathcal{O}}, \tag{10}$$

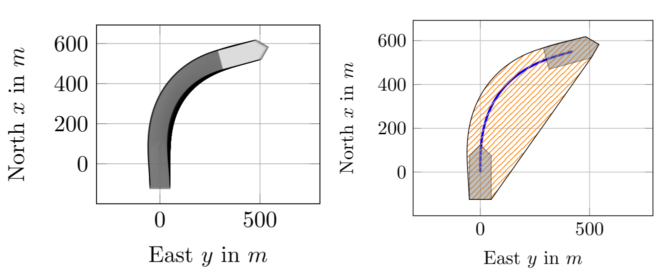

Fig. 4: Comparison of a multitude of single polygons (at every time step) and a single swept hull (orange), where the polygon of the vessel at the start and at the end is gray and the trajectory of the vessel is blue.

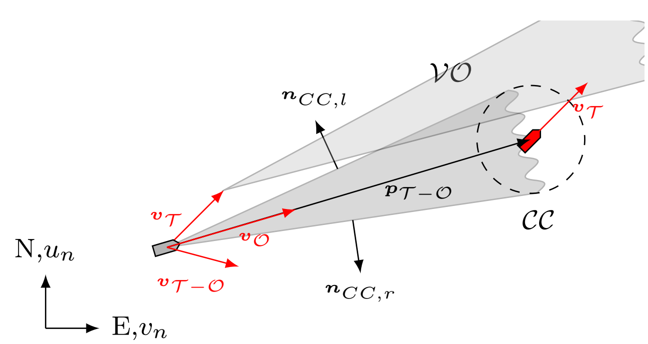

Fig. 5: Encounter of the ownship and target in the inertial frame and in the velocity frame.

which means for the implementation using the CC that

$$\boldsymbol{n}_{CC,l}^{\top}(\boldsymbol{v}_{\mathcal{O}}^{*} - \boldsymbol{v}_{\mathcal{T}}) \geq 0 \wedge \boldsymbol{n}_{CC,r}^{\top}(\boldsymbol{v}_{\mathcal{O}}^{*} - \boldsymbol{v}_{\mathcal{T}}) \geq 0, \qquad (11)$$

where $\boldsymbol{v}_{\mathcal{O}}^{*}(t)$ is the inertial velocities stemming from a MP. The logic of finding the correct COLREG situation is task of a COLREG-analyzer system. The interpretation of the COLREG states is presented here:

- Overtake: Both, the port- and starboard normal vectors are used to define the CC. As the vessel is allowed to overtake on both sides.
- Crossing: Only the starboard halfplane is considered, as the ownship is obliged to avoid the target by this side.
- Crossing-stand-on and being-overtaken: The database is reduced to only straight maneuvers.

V. Simulation Results

Verification of the presented unified approach to COLREG conformal trajectory generation is done in simulation. At first, the general path following behavior is analyzed. In Fig. 6 a long passage out of the Kiel Fjord is planned, which is a $4.7\,\mathrm{km}$ long journey. As a model a Nomoto class vessel (5) is used, with parameters of a ferry of $L_{pp} = 60\,\mathrm{m}$ length. The algorithm easily follows the waypoints while still respecting the vessel dynamics and terminates when it reaches the last waypoint.
Now, COLREG situations can be constructed and tested against the algorithm. The quality of the solution is measured by reaching the destination within a given distance, the fulfillment of the COLREG compliance and amount of course changing maneuvers. In Fig. 7 a scenario is presented, where a target vessel is crossing the ownship's path from the inlet of a river. Simultaneously, another target approaches head-on, but is not pressuring the ownship to take avoiding action. The same database as for the example in Fig. 6 is used. Instead of terminating the algorithm when it reaches the last waypoints, it is terminated when the tree reaches a cardinality of 500 nodes. After the ownship heads for the first waypoint, the crossing target vessel is detected. The ownship performs a course change to starboard side, as it has to give way to the target vessel. After the target vessel has passed, the ownship has to avoid the border of coastline, so it performs a more aggressive maneuver to head portside back to the track.

VI. Conclusions

As visible from Fig. 7, the presented approach is capable of generating COLREG-compliant trajectories. The solution is based on a set of precalculated MPs, which are used to reduce the numeric complexity of the problem. The solution consists of full state and input discretization, which means that the solution can directly be used for, e.g., a tracking controller. This approach brings a lot of degrees of freedom from an implementation standpoint. For example, the design of the database is not limited to what is presented in Sec. IV-B. The database can be extended to include more complex maneuvers, like a full turn, constant turn rate, or also to include less maneuvers until more MPs are needed. The main task here is to find the trade-off between number of MPs in

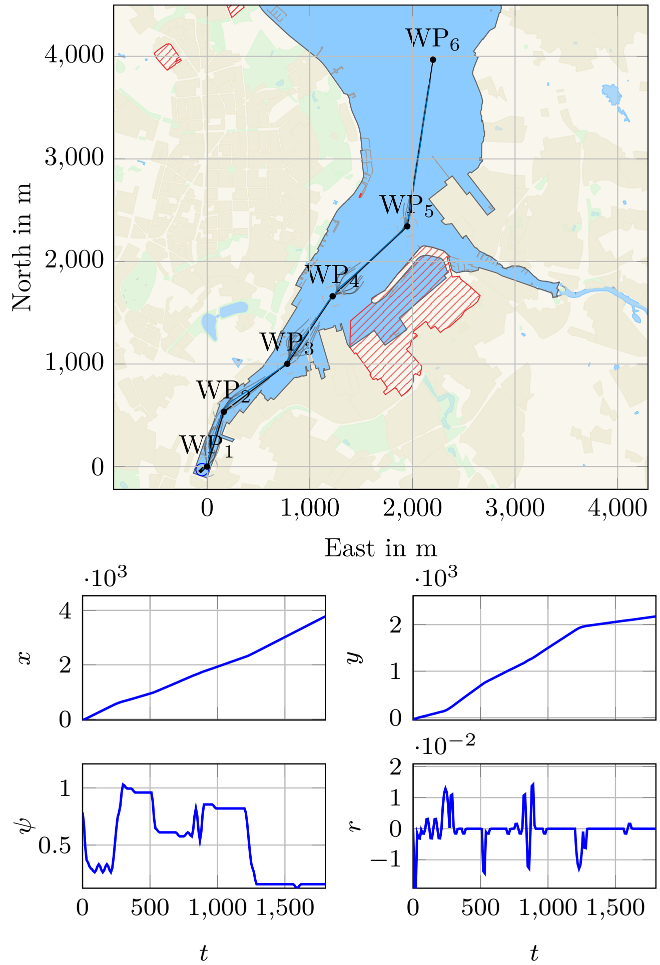

Fig. 6: Test maneuver for the RRT-algorithm in combination with motion primitives. The given waypoints are black. The resulting trajectory is blue, the branches of the calculated tree are given in gray color. Below are the associated state trajectories.

the database that have to be searched for validity, and the goal to approximate the full dynamics of the vessel as good as possible.
For the future, aspects of the RRT* are of great interest, but difficult to implement. In general, the RRT* aims to optimize the tree on run-time by reconnecting the nodes, which is demanding to combine with the explicit time dependency of COLREG-conformal maneuvers. When trying to reconnect nodes, the trajectory has to be checked again for COLREG-compliance. Furthermore, COLREG rules are only defined for one-to-one encounters of vessels. While this is a rather common case for offshore situations, in regions of high marine traffic, one-to-many scenarios are of interest.

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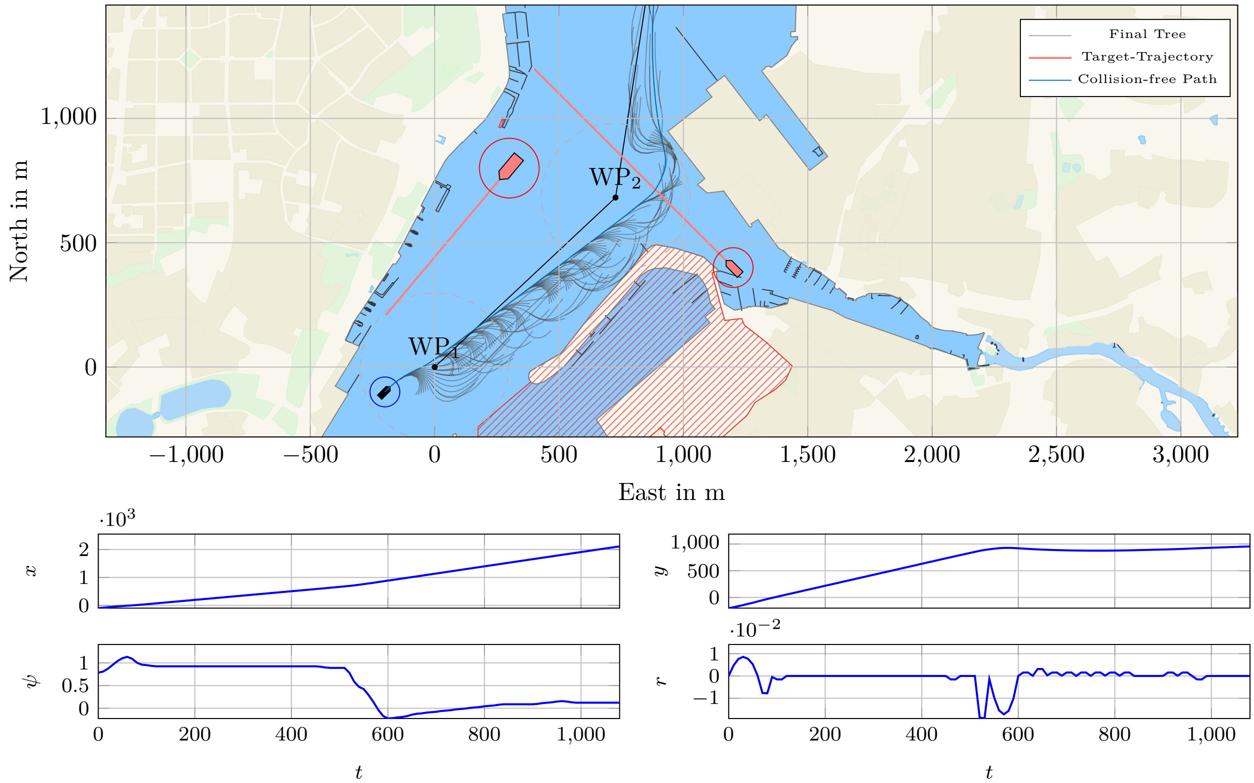

Fig. 7: The charts of the ownship's (black) collision avoidance maneuver are on top where the tree (gray) is, the obstacle (orange) and the targets (red), together with their predicted trajectories. Below are the associated state trajectories of the ownship.

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